

AI Engineer Career Roadmap

Student • Python & Data Analysis • Timeline: 3 Years • Goal: AI Engineer

Chain-of-Thought Planning

Current Position

- Status: Student
- Strengths: Python, Data Analysis

Skill Area	Current Level	Required Level	Priority	Est. Time
Machine Learning	■■■■■	■■■■■	High	4 months
Deep Learning	■■■■■	■■■■■	High	4 months
LLMs & GenAI	■■■■■	■■■■■	High	5 months
MLOps / Cloud	■■■■■	■■■■■	Medium	4 months
SQL & Data Eng.	■■■■■	■■■■■	Low	1 month
System Design	■■■■■	■■■■■	Medium	3 months

Target Goal

- Role: AI Engineer
- Skills Needed: ML, DL, MLOps, LLMs
- Target Salary: \$130k–\$180k+
- Timeline: 3 Years

Recommended Learning Plan

Year 1 — Foundations

- ML with scikit-learn (Coursera / fast.ai)
- Deep Learning: PyTorch / TensorFlow
- Math: Linear Algebra, Stats refresher

Year 2 — Specialization

- LLMs: Hugging Face, LangChain, RAG
- Cloud AI: AWS SageMaker / GCP Vertex AI
- MLOps: Docker, Airflow, CI/CD pipelines

Year 3 — Professional Edge

Platform / Action	Goal	Frequency
LinkedIn — post projects	Build public profile & credibility	2× / month
Kaggle / GitHub activity	Showcase code & problem-solving	Weekly
AI Twitter / X community	Follow researchers, join discussions	Daily 10 min
Hackathons & Meetups	Real connections + team projects	1 per quarter
Cold email / DM engineers	Mentorship & referrals	Monthly

Suggested Projects

Year 1 Projects

- Kaggle competition (Top 30%)
- EDA + ML pipeline on real dataset
- GitHub portfolio setup

Year 2 Projects

- Fine-tune an LLM (LoRA / QLoRA)
- RAG chatbot with vector DB
- Deploy model API on AWS/GCP

Year 3 Projects

Year-by-Year Milestones

Year 1 (Months 1–12)

- ✓ Complete ML + DL courses
- ✓ First Kaggle top-30%
- ✓ 5 GitHub projects live
- ✓ Active LinkedIn profile

Year 2 (Months 13–24)

- ✓ LLM fine-tuning project
- ✓ Cloud deployment live
- ✓ MLOps pipeline built
- ✓ 500+ LinkedIn followers

Year 3 (Months 25–36)

- ✓ Apply to AI Engineer roles
- ✓ Open-source contributions
- ✓ Technical blog (5+ posts)
- ✓ Land AI Engineer job

Immediate Next Actions (This Week)

- Install PyTorch & complete fast.ai Lesson 1
- Set up GitHub profile — push your first ML notebook
- Create LinkedIn & write your 'Student → AI Engineer' story post
- Join fast.ai & Hugging Face Discord communities
- Pick one Kaggle beginner competition and submit by Day 14
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